

CS 70 — Discussion 1D

Problem 1: Well-Ordering Principle

In this problem we derive the Well-Ordering Principle (WOP) from strong induction. The WOP states: for every non-empty subset $S \subseteq \mathbb{N}$, there exists a smallest element $x \in S$ such that $\forall y \in S, x \leq y$.

Part (a): Significance of S being non-empty

If S is allowed to be empty, the WOP fails trivially: the empty set has no elements, so no smallest element can be produced. The non-empty condition guarantees a witness $z \in S$ exists, which is necessary to begin the induction argument.

Part (b): Induction variable

The induction should be based on z , the natural number guaranteed to exist in S by non-emptiness. We induct on z to show every non-empty subset containing an element $\leq z$ has a least element.

Part (c): Induction hypothesis

P(z): For every non-empty subset $S \subseteq \mathbb{N}$, if $z \in S$, then S has a least element; i.e., $\exists x \in S$ such that $\forall y \in S, x \leq y$.

Part (d): Base case

The base case is $z = 0$, verified for all non-empty subsets S containing 0. Since 0 is the smallest natural number, 0 must be the least element of any such S . The base case holds for every non-empty $S \subseteq \mathbb{N}$ with $0 \in S$. ✓

Part (e): Inductive step (strong induction)

Strong inductive hypothesis: Assume $P(m)$ holds for all $m \leq z$.

Goal: Show $P(z+1)$: if $z+1 \in S$, then S has a least element.

We split into two cases:

Case 1: S contains some element $m \leq z$. By the strong inductive hypothesis, $P(m)$ holds, so S has a least element.

Case 2: S contains no element in $\{0, 1, \dots, z\}$. Then $z + 1$ is smaller than every other element of S , so $z + 1$ is the least element of S .

In both cases S has a least element, completing the induction. $\square$

Part (f): Modification for simple induction

To make the proof work by simple induction, we strengthen the induction hypothesis to:

P(z): Every non-empty subset of $\{0, 1, \dots, z\}$ has a least element.

In the inductive step, given a non-empty S containing $z+1$: if $S \cap \{0, \dots, z\}$ is non-empty, apply $P(z)$ to that restricted set to find the least element of S ; otherwise $z + 1$ is the least element. Only the previous step $P(z)$ is needed, not all prior steps.

Problem 2: Graph Basics

Consider the directed graph G with vertices $\{A, B, C, D, E, F, G, H\}$ and edges as shown.

Part (a): Vertex and edge sets

$$V = \{A, B, C, D, E, F, G, H\}$$

$$E = \{(A, B), (A, F), (B, C), (B, E), (C, D), (D, B), (D, H), (E, G), (E, D), (H, G), (F, G), (G, F), (F, E)\}$$

Part (b): In-degrees and out-degrees

- Highest in-degree: G with in-degree 3.
- Lowest in-degree: A with in-degree 0.
- Vertices with equal in-degree and out-degree: B (2,2), E (2,2), F (2,2), H (1,1), C (1,1).

Part (c): Paths from B to F

- $B \rightarrow E \rightarrow G \rightarrow F$ (shortest path, length 3)
- $B \rightarrow E \rightarrow D \rightarrow H \rightarrow G \rightarrow F$
- $B \rightarrow C \rightarrow D \rightarrow H \rightarrow G \rightarrow F$

Part (d): Cycles

- $\{(B, C), (C, D), (D, B)\}$ — **Yes**, this is a cycle.
- $\{(F, G), (G, F)\}$ — **Yes**, this is a cycle.
- $\{(A, B), (B, C), (C, D), (D, B)\}$ — **No**, does not return to A .
- $\{(B, C), (C, D), (D, H), (H, G), (G, F), (F, E), (E, D), (D, B)\}$ — **No**, vertex D is visited twice.

Part (e): Walks

A walk is any sequence of edges with no restrictions on vertex or edge revisiting.

- i. $\{(E, G)\}$ — **Yes**, edge $E \rightarrow G$ exists.
- ii. $\{(E, G), (G, F)\}$ — **No**, edge $G \rightarrow F$ does not exist (only $F \rightarrow G$).
- iii. $\{(F, G), (G, F)\}$ — **Yes**.
- iv. $\{(A, B), (B, C), (C, D)\}$ — **No**, edge $C \rightarrow D$ does not exist.
- v. $\{(E, G), (G, F), (F, G), (G, F)\}$ — **Yes**.
- vi. $\{(E, D), (D, B), (B, E), (E, D), (D, H), (H, G), (G, F)\}$ — **Yes**.

Part (f): Tours

A tour is a walk that visits every edge exactly once.

- i. $\{(E, G)\}$ — **No**, does not visit every edge.
- ii. $\{(E, G), (G, F)\}$ — **No**, does not visit every edge.
- iii. $\{(F, G), (G, F)\}$ — **Yes**, visits both edges of the cycle exactly once.
- iv. $\{(E, D), (D, B), (B, E), (E, D), (D, H), (H, G), (G, F)\}$ — **No**, edge (E, D) is visited twice.

Part (g): True/False — degree at most 1 implies no cycle

True. A cycle requires every vertex in the cycle to have degree at least 2. If every vertex has degree at most 1, no cycle can exist.

Part (h): True/False — degree at least 2 implies a cycle exists

True. Start at any vertex and follow edges. Since every vertex has degree ≥ 2 , whenever you arrive at a vertex there is always another edge to leave on. Since the graph is finite, you must eventually revisit a vertex, forming a cycle.

Part (i): True/False — degree at most 2 implies not connected

False. A triangle $(A-B-C-A)$ is connected and every vertex has degree exactly 2.

Problem 3: Eulerian Tour and Eulerian Walk

1. **Is there an Eulerian tour?** No. An Eulerian tour requires every vertex to have even degree. The graph has vertices of odd degree, so no Eulerian tour exists.
2. **Is there an Eulerian walk?** Yes. The graph has exactly 2 vertices of odd degree. An Eulerian walk exists starting at one odd-degree vertex and ending at the other.
3. **Condition for an Eulerian walk in an undirected graph:** The graph is connected and has exactly 0 or 2 vertices of odd degree. (0 odd-degree vertices gives an Eulerian tour; exactly 2 gives an Eulerian walk between the two odd-degree vertices.)

Problem 4: Odd Degree Vertices

Claim: In any undirected graph $G = (V, E)$, the number of vertices with odd degree is even.

Part (i): Direct proof

By the handshaking lemma, $\sum_{v \in V} \deg(v) = 2|E|$, which is even. Partition V into O (odd-degree vertices) and E (even-degree vertices):

$$\sum_{v \in O} \deg(v) + \sum_{v \in E} \deg(v) = 2|E|$$

The sum over E is even (sum of even numbers). Since the total is even, $\sum_{v \in O} \deg(v)$ must also be even. A sum of odd numbers is even if and only if there are an even number of them. Therefore $|O|$ is even. $\square$

Part (ii): Induction on $m = |E|$

Base case. $m = 0$: the graph has no edges, all vertices have degree 0 (even), so 0 odd-degree vertices. $\checkmark$

Inductive hypothesis. Assume any graph with $m = n$ edges has an even number of odd-degree vertices.

Inductive step. Consider a graph G with $n + 1$ edges. Remove one edge (u, v) to obtain G' with n edges. By the IH, G' has an even number of odd-degree vertices. Now add the edge back. There are three cases:

- *Both u and v have even degree in G' :* adding the edge makes both odd. The count increases by 2 — still even.
- *Both u and v have odd degree in G' :* adding the edge makes both even. The count decreases by 2 — still even.
- *One of u, v is odd, the other even:* adding the edge flips both. The count is unchanged — still even.

In all cases the count remains even for $m = n + 1$. $\square$

Part (iii): Induction on $n = |V|$

Base case. $n = 1$: one vertex, no edges, degree 0, so 0 odd-degree vertices. ✓

Inductive hypothesis. Assume any graph with n vertices has an even number of odd-degree vertices.

Inductive step. Add the $(n + 1)$ th vertex v with k edges connecting it to existing vertices. Two cases:

- *k is even:* v has even degree, and an even number of existing vertices have their parity flipped. The count of odd-degree vertices does not change — still even.
- *k is odd:* v has odd degree (+1 to count), and an odd number of existing vertices have their parity flipped (net change of odd many). Total change: odd +1 = even. Still even.

In both cases the count remains even for $n + 1$ vertices. □

Part (iv): Well-Ordering Principle

Assume for contradiction that there exists a graph where the number of odd-degree vertices is odd. Let $S = \{m \in \mathbb{N} \mid \text{there exists a graph with } m \text{ edges where the number of odd-degree vertices is odd}\}$. By assumption S is non-empty, so by the WOP S has a smallest element m_0 .

First, $m_0 > 0$: a graph with 0 edges has all vertices with degree 0 (even), giving 0 odd-degree vertices, which is even — so $0 \notin S$.

Let G be a graph with m_0 edges and an odd number of odd-degree vertices. Remove one edge (u, v) to get G' with $m_0 - 1$ edges. Since m_0 is minimal, G' has an even number of odd-degree vertices.

Now add the edge (u, v) back. By the same case analysis as in part (ii), adding one edge changes the odd-degree count by 0 or ± 2 . Therefore if G' has an even count, G must also have an even count — contradicting our assumption that G has an odd count.

Therefore S is empty and the claim holds for all graphs. □